

DBMS – PAYMENT TRANSACTION MANAGEMENT SYSTEM

Project Title: EcoCart – Sustainable Products E-Commerce Order Management System

Skill Task: DBMS

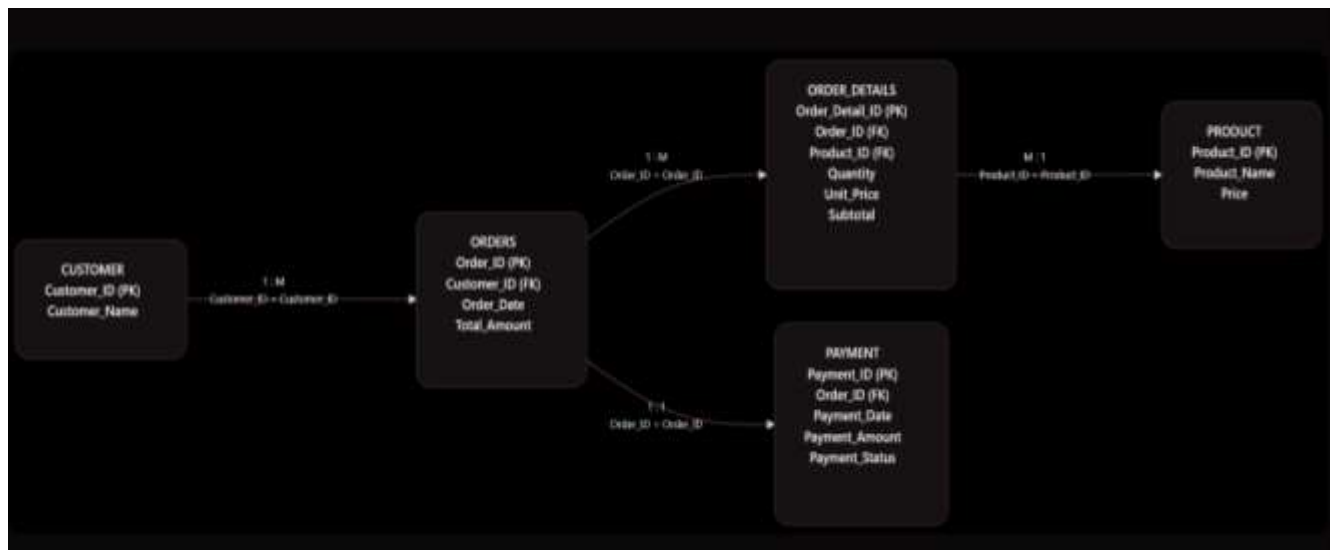
Week: 5

1. Database Design

1.1 Payment Table

Attribute	Data Type	Constraint
<i>Payment_ID</i>	INT	Primary Key
<i>Order_ID</i>	INT	Foreign Key
<i>Payment_Mode</i>	VARCHAR2(20)	NOT NULL
<i>Payment_Date</i>	DATE	NOT NULL
<i>Payment_Amount</i>	NUMBER(10,2)	NOT NULL
<i>Payment_Status</i>	VARCHAR2(20)	NOT NULL

2. Table Relationships



3. Table Creation

3.1 Create Payment Table

```
SQL> CREATE TABLE Payment (  
2     Payment_ID      NUMBER(5) PRIMARY KEY,  
3     Order_ID        NUMBER(5),  
4     Payment_Mode     VARCHAR2(20) NOT NULL,  
5     Payment_Date     DATE NOT NULL,  
6     Payment_Amount   NUMBER(10,2) NOT NULL,  
7     Payment_Status   VARCHAR2(20) NOT NULL,  
8     CONSTRAINT fk_payment_order  
9     FOREIGN KEY (Order_ID)  
10    REFERENCES Orders(Order_ID)  
11 );
```

Table created.

4. Payment Data Insertion

```
SQL> INSERT INTO Payment  
2 (Payment_ID, Order_ID, Payment_Mode, Payment_Date, Payment_Amount, Payment_Status)  
3 VALUES  
4 (501, 201, 'UPI', SYSDATE, 2500.00, 'Successful');
```

1 row created.

```
SQL> INSERT INTO Payment  
2 (Payment_ID, Order_ID, Payment_Mode, Payment_Date, Payment_Amount, Payment_Status)  
3 VALUES  
4 (502, 202, 'Card', SYSDATE, 1800.00, 'Successful');
```

1 row created.

```
SQL> INSERT INTO Payment  
2 (Payment_ID, Order_ID, Payment_Mode, Payment_Date, Payment_Amount, Payment_Status)  
3 VALUES  
4 (503, 203, 'Net Banking', SYSDATE, 3200.00, 'Failed');
```

1 row created.

```
SQL> INSERT INTO Payment  
2 (Payment_ID, Order_ID, Payment_Mode, Payment_Date, Payment_Amount, Payment_Status)  
3 VALUES  
4 (504, 204, 'UPI', SYSDATE, 1500.00, 'Successful');
```

1 row created.

5. Manage Payment Transactions

5.1 Display Successful Payments

```
SQL> SELECT *
      2 FROM Payment
      3 WHERE Payment_Status = 'Successful';
```

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT
501	201	UPI	16-SEP-26	2500
502	202	Card	16-SEP-26	1800
504	204	UPI	16-SEP-26	1500

5.2 Display Failed Payments

```
SQL> SELECT *
      2 FROM Payment
      3 WHERE Payment_Status = 'Failed';
```

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT
503	203	Net Banking	16-SEP-26	3200

6. Modify Payment Status

```
SQL> UPDATE Payment
  2 SET Payment_Status = 'Successful'
  3 WHERE Payment_ID = 503;

1 row updated.

SQL> SELECT *
  2 FROM Payment
  3 WHERE Payment_ID = 503;
```

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT
503	203	Net Banking	16-SEP-26	3200
Successful				

7. Analyze Payment Methods

```
SQL> SELECT
  2 Payment_Mode,
  3 COUNT(*) AS Total_Transactions
  4 FROM Payment
  5 GROUP BY Payment_Mode;
```

PAYMENT_MODE	TOTAL_TRANSACTIONS
Net Banking	1
Card	1
UPI	2

Calculate amount collected by payment mode

```
SQL> SELECT
  2 Payment_Mode,
  3 SUM(Payment_Amount) AS Total_Amount
  4 FROM Payment
  5 WHERE Payment_Status = 'Successful'
  6 GROUP BY Payment_Mode;
```

PAYMENT_MODE	TOTAL_AMOUNT
Net Banking	3200
Card	1800
UPI	4000

8. Payment Transaction Report

```
SQL> SELECT
  2     p.Payment_ID,
  3     o.Order_ID,
  4     c.Customer_ID,
  5     c.Customer_Name,
  6     p.Payment_Mode,
  7     p.Payment_Date,
  8     p.Payment_Amount,
  9     p.Payment_Status
 10 FROM Payment p
 11 JOIN Orders o
 12     ON p.Order_ID = o.Order_ID
 13 JOIN Customer c
 14     ON o.Customer_ID = c.Customer_ID
 15 ORDER BY p.Payment_Date DESC;
```

PAYMENT_ID	ORDER_ID	CUSTOMER_ID	CUSTOMER_NAME	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT	PAYMENT_STATUS
504	204	104	Karthik S	UPI	16-SEP-26	1500	Successful
503	203	103	Priya	Net Banking	16-SEP-26	3200	Successful

PAYMENT_ID	ORDER_ID	CUSTOMER_ID	CUSTOMER_NAME	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT	PAYMENT_STATUS
502	202	102	Rahul	Card	16-SEP-26	1800	Successful
501	201	101	Anitha Kumar				

PAYMENT_ID	ORDER_ID	CUSTOMER_ID	CUSTOMER_NAME	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT	PAYMENT_STATUS
				UPI	16-SEP-26	2500	Successful